



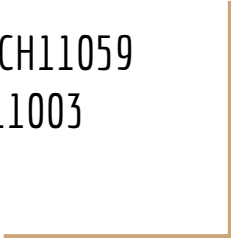
AI4005- Project Presentation

ANN-parameterized GMM Clustering Algorithm

P1: Understanding GMMs

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Agenda

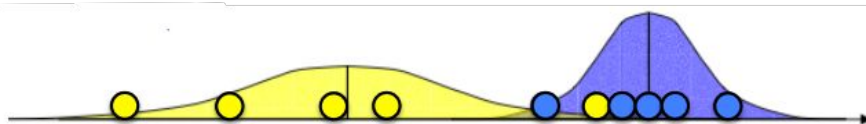
- Understand Gaussian Mixture Models
- Expectation Maximization Algorithm
- Visualization
- Time Complexity Analysis
 - Potential Culprits
 - Mitigation Strategy

Mixture Models

- Clustering Methods:
 - Hard Clustering:
 - Clusters do not overlap.
 - Each element either belongs to a cluster or does not.
 - Soft Clustering:
 - Clusters may overlap.
 - Strength of association between clusters and instances.
- Mixture Models:
 - Probabilistically grounded way of doing soft clustering.
 - Each cluster is represented by a generative model (e.g., Gaussian or multinomial).
 - Parameters (e.g., mean/covariance) are unknown.
- Expectation Maximization (EM) Algorithm:
 - Automatically discovers all parameters for the K "sources".

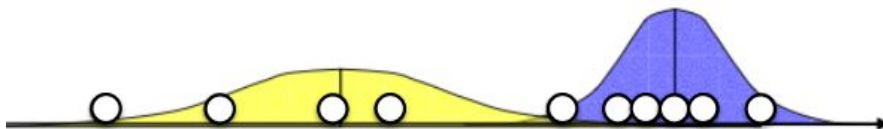
Mixture Models in 1-D

- Observations $x_1, \dots, x_n$
 - $K=2$ Gaussians with unknown μ, σ^2
 - Estimation is trivial if we know the source of each observation



$$\mu_b = \frac{x_1 + x_2 + \dots + x_{n_b}}{n_b}$$
$$\sigma_b^2 = \frac{(x_1 - \mu_b)^2 + \dots + (x_{n_b} - \mu_b)^2}{n_b}$$

- What if we don't know the source?
- If we knew parameters of the Gaussians (μ, σ^2)
 - Can guess whether point is more likely to be a or b



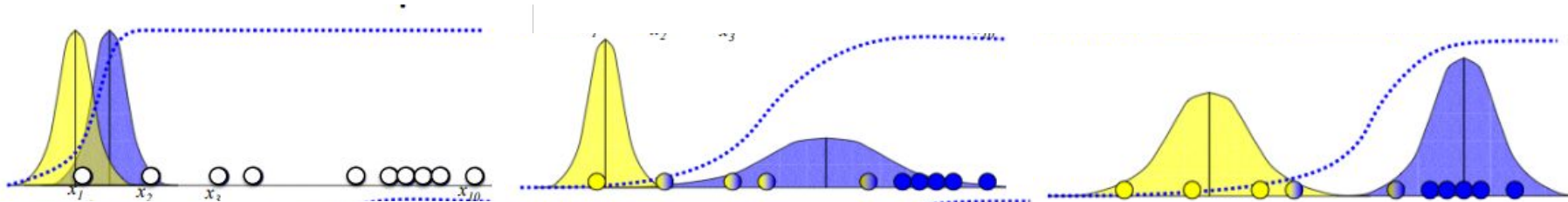
$$P(b | x_i) = \frac{P(x_i | b)P(b)}{P(x_i | b)P(b) + P(x_i | a)P(a)}$$

$$P(x_i | b) = \frac{1}{\sqrt{2\pi\sigma_b^2}} \exp\left\{-\frac{(x_i - \mu_b)^2}{2\sigma_b^2}\right\}$$

Expectation Maximization

- Chicken and egg problem:
 - Need (μ_a, σ_a^2) and (μ_b, σ_b^2) to guess source of points
 - Need to know source to estimate (μ_a, σ_a^2) and (μ_b, σ_b^2)
- EM algorithm:
 - Start with two randomly placed Gaussians (μ_a, σ_a^2) , (μ_b, σ_b^2)
 - *E-Step*: For each point: $P(b | x_i)$ = Does it look like it came from b?
 - *M-Step*: Adjust (μ_a, σ_a^2) and (μ_b, σ_b^2) to fit points assigned to them
 - Iterate until convergence

Visualization (1-D)



$$P(x_i | b) = \frac{1}{\sqrt{2\pi\sigma_b^2}} \exp\left\{-\frac{(x_i - \mu_b)^2}{2\sigma_b^2}\right\}$$

$$b_i = P(b | x_i) = \frac{P(x_i | b)P(b)}{P(x_i | b)P(b) + P(x_i | a)P(a)}$$

$$a_i = P(a | x_i) = 1 - b_i$$

$$\mu_b = \frac{b_1x_1 + b_2x_2 + \dots + b_nx_n}{b_1 + b_2 + \dots + b_n}$$

$$\sigma_b^2 = \frac{b_1(x_1 - \mu_b)^2 + \dots + b_n(x_n - \mu_b)^2}{b_1 + b_2 + \dots + b_n}$$

$$\mu_a = \frac{a_1x_1 + a_2x_2 + \dots + a_nx_n}{a_1 + a_2 + \dots + a_n}$$

$$\sigma_a^2 = \frac{a_1(x_1 - \mu_a)^2 + \dots + a_n(x_n - \mu_a)^2}{a_1 + a_2 + \dots + a_n}$$

$$P(b) = \frac{b_1 + b_2 + \dots + b_n}{n}$$

$$P(a) = 1 - P(b)$$

Visualization (multi-dimensional)

- Data with attributes, from Gaussian sources $c_1 \dots c_k$

- How typical is x_i under source c ?

$$P(\vec{x}_i | c) = \frac{1}{\sqrt{2\pi|\Sigma_c|}} \exp\left\{-\frac{1}{2}(\vec{x}_i - \vec{\mu}_c)^T \Sigma_c^{-1} (\vec{x}_i - \vec{\mu}_c)\right\}$$

$$\sum_a \sum_b (x_{ia} - \mu_{ca}) [\Sigma_c^{-1}]_{ab} (x_{ib} - \mu_{cb})$$

- How likely that x_i came from c ?

$$P(c | \vec{x}_i) = \frac{P(\vec{x}_i | c)P(c)}{\sum_{c=1}^k P(\vec{x}_i | c)P(c)}$$

- How important is x_i for source c :

$$w_{i,c} = P(c | \vec{x}_i) / (P(c | \vec{x}_1) + \dots + P(c | \vec{x}_n))$$

- Mean of attribute a in items assigned to c

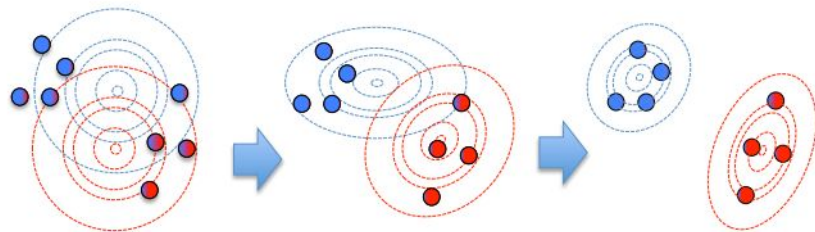
$$\mu_{ca} = w_{c1}x_{1a} + \dots + w_{cn}x_{na}$$

- Covariance of a and b in items from c

$$\Sigma_{cab} = \sum_{i=1}^n w_{ci} (x_{ia} - \mu_{ca})(x_{ib} - \mu_{cb})$$

- Prior: how many items assigned to c

$$P(c) = \frac{1}{n} (P(c | \vec{x}_1) + \dots + P(c | \vec{x}_n))$$



How to pick K?

- Probabilistic model:
 - Tries to "fit" the data (maximize likelihood)
- Pick K that makes L as large as possible?
 - $K = n$: each data point has its own "source"
 - May not work well for new data points
- Split points into training set T and validation set V
 - For each K: fit parameters of T, measure likelihood of V
 - Sometimes still best when $K = n$
- Occam's razor: pick "simplest" of all models that fit
 - Bayes Information Criterion (BIC): $\max \{L - \frac{1}{2}p \log n\}$
 - Akaike Information Criterion (AIC): $\min \{2p - \bar{L}\}$

$$L = \log P(x_1 \dots x_n) = \sum_{i=1}^n \log \sum_{k=1}^K P(x_i | k) P(k)$$

L ... likelihood, how well
our model fits the data
p ... number of parameters
how "simple" is the model

Quick Summary so far...

- Walked through 1D version:
 - Works for higher dimensions
 - d-dimensional Gaussians, can be non-spherical
 - Works for discrete data (text)
 - d-dimensional multinomial distributions (pLSI)
- Maximizes likelihood of the data:
- Similar to K-means
 - Sensitive to starting point, converges to a local maximum
 - Convergence: when change in $P(x_1 \dots x_n)$ is sufficiently small
 - Cannot discover K (likelihood keeps growing with K)
- Different from K-means
 - Soft clustering: instance can come from multiple "clusters"
 - Covariance: notion of "distance" changes over time

$$P(x_1 \dots x_n) = \prod_{i=1}^n \sum_{k=1}^K P(x_i | k) P(k)$$

Time Complexity Analysis

Algorithm 1 Gaussian Mixture Model (GMM)

```
1: Input: Data  $\{x_1, x_2, \dots, x_N\}$ , number of Gaussian components  $K$ 
2: Initialize: Randomly initialize parameters for  $K$  Gaussian components:
    $\{\mu_k, \Sigma_k, \pi_k\}$ 
3: repeat
4:   Expectation Step:
5:   for  $i = 1$  to  $N$  do
6:     for  $k = 1$  to  $K$  do
7:       Compute posterior probability (responsibility) of component  $k$  for
       data point  $x_i$ :
```

$$\gamma_{ik} = \frac{\pi_k \mathcal{N}(x_i | \mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(x_i | \mu_j, \Sigma_j)}$$

```
8:     end for
9:   end for
10:  Maximization Step:
11:  for  $k = 1$  to  $K$  do
12:    Update parameters for component  $k$ :
```

$$\mu_k = \frac{\sum_{i=1}^N \gamma_{ik} x_i}{\sum_{i=1}^N \gamma_{ik}}, \quad \Sigma_k = \frac{\sum_{i=1}^N \gamma_{ik} (x_i - \mu_k)(x_i - \mu_k)^T}{\sum_{i=1}^N \gamma_{ik}}, \quad \pi_k = \frac{1}{N} \sum_{i=1}^N \gamma_{ik}$$

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13:  end for
14: until Convergence
15: Output: Parameters  $\{\mu_k, \Sigma_k, \pi_k\}$  for  $K$  Gaussian components
```

Initialization Step: $O(Kd^2)$

Expectation (E) Step: $O(NKd^3)$

Maximization (M) Step: $O(K(Nd^2))$

Total Time Complexity:

$$O(T(Kd^2 + NKd^3 + K(Nd^2))) \approx O(TNKd^3)$$

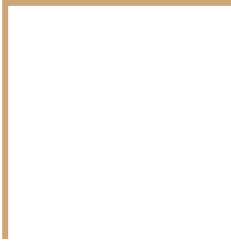
- N is the number of data points
- K is the number of clusters
- d is the dimensionality of each data point

Our Plan

- Reduce time complexity of the base GMM formulation
 - [A Fast Incremental Gaussian Mixture Model](#) promises to reduce the time complexity of vanilla-GMM to $\overline{O(NKd^2)}$
 - Uses Mahalanobis distance $d_M^2(\mathbf{x}, j) = (\mathbf{x} - \boldsymbol{\mu}_j)^T \mathbf{C}_j^{-1} (\mathbf{x} - \boldsymbol{\mu}_j)$
 - Joint Learning of Unsupervised Dimensionality Reduction to reduce 'd'.
- Improve the ANN-parameterized GMM formulation
 - Current Code: [ANN parameterized GMM](#)
 - Predicts mean vector and covariance matrix (assumed diagonal).
 - It should be extended for any general covariance matrix.
 - Add prediction of prior in the ANN output
 - Cholesky Decomposition for numerical stability
 - Integrate with Fast Incremental GMM
 - Use Genetic Algorithm to look through the search space to find the optimum ANN architecture

References

1. [Gaussian Mixture Models Clustering Algorithm Explained by Cory Maklin](#)
2. [Expectation Maximization Algorithm](#)
3. [GMM Lecture Notes by Victor Lavrenko](#)



Thank You

Any questions?

